

D286

Java Fundamentals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A developer on a payroll team that ships every two weeks writes `void raise(int salary) { salary = salary + 5000; }`. In main, the code runs `int pay = 60000; raise(pay); System.out.println(pay);`. The team lead notes the code passed peer review. What prints?

- A. It will not compile, because a parameter can never be reassigned.
- B. 60000
- C. 5000
- D. 65000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A method `void birthday(Person p) { p.setAge(p.getAge() + 1); }` is called with a `Person` object whose age is 30. After the call, what does the caller's `person.getAge()` return?

- A. 31
- B. 1
- C. 30
- D. It cannot be known, because Java copies the whole object into p.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An inventory app splits 17 pallets evenly across 4 trucks with `int perTruck = 17 / 4;` State rules let each truck carry up to 20 pallets. What value is stored in `perTruck`?

- A. 20
- B. 4.25
- C. 5
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Given `int total = 9` and `int count = 2`, which expression makes `double avg` equal 4.5?

- A.** `Math.round(total / count)`
- B.** `(double) (total / count)`
- C.** `(double) total / count`
- D.** `total / count * 1.0`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A login service written by an outside contractor checks if `(inputRole == storedRole)`. Both variables hold the text `ADMIN`, but `inputRole` was built at runtime from keyboard input. The service handles about 3,000 logins a day. Why does the check return false?

- A.** The daily login volume overflows the String pool.
- B.** `==` always returns false for any two Strings.
- C.** `==` compares object references, not text; use `inputRole.equals(storedRole)`.
- D.** String comparison is case sensitive, so `ADMIN` does not match.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which expression correctly tests whether two `int` primitives `a` and `b` hold the same value?

- A.** `a = b`
- B.** `a.equals(b)`
- C.** `a == b`
- D.** `Integer.compare(a, b) == 1`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A billing class already has `double total(double price, int qty)`. Working against a Friday release, a developer adds `int total(double price, int qty)` to the same class. What happens?

- A.** The new method overrides the old one.
- B.** The class fails to compile, because return type alone cannot distinguish overloads.
- C.** Two methods with the same name are never allowed in one class.
- D.** It compiles, and Java picks the method by the variable receiving the result.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A Ticket class declares private static int issued = 0; and its constructor runs issued++. A theater box office that has been open since 1998 creates three Ticket objects for a Sunday matinee. What is the value of issued afterward?

- A.** 3
- B.** 1 in each Ticket object
- C.** 1998
- D.** 0, because static fields cannot change after start-up

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A BankAccount class stores balance as a public double field. The security team, which reviews code on Tuesdays, flags that other classes can set balance to a negative number. Which change best applies encapsulation?

- A.** Make balance private and add deposit and withdraw methods that reject invalid amounts.
- B.** Change balance to protected.
- C.** Make balance final so no one can set it wrong.
- D.** Keep it public but add a comment asking callers not to go negative.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Class Shape has double area() that returns 0. Class Circle extends Shape and overrides area() to return $3.14 * r * r$. A drawing tool used by 400 architects runs Shape s = new Circle(2); System.out.println(s.area()); What prints?

- A.** 6.28
- B.** It does not compile, because s is declared as a Shape.
- C.** 0.0
- D.** 12.56

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Shape s = new Circle(2); is followed by s.getRadius(); where getRadius() is defined only in Circle, not in Shape. What happens?

- A.** It returns 2, because the object is really a Circle.
- B.** It throws a ClassCastException at runtime.
- C.** It returns 0, the default for a missing method.
- D.** A compile error, because Shape has no getRadius() method.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A Product class written for a grocery chain with 60 stores declares exactly one constructor, public Product(String sku, double price). Another class calls new Product(). What happens?

- A.** It compiles but throws a NullPointerException at runtime.
- B.** A compile error, because no no-argument constructor exists.
- C.** It creates a Product with a null sku and a 0.0 price.
- D.** It works, because Java always supplies a no-argument constructor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A developer ~~on a payroll team that ships every two weeks~~ writes `void raise(int salary) { salary = salary + 5000; }`. In main, the code runs `int pay = 60000; raise(pay); System.out.println(pay);` ~~The team lead notes the code passed peer review.~~ What prints?

Target word: value of the caller's variable **Decides:** The int is copied into salary; changing salary does not change pay.

A	Absolute language	Non-final parameters can be reassigned; the change just stays local.
B	Correct	Java passes a copy of the int; pay is untouched.
C	Wrong stem number	The raise amount, not the value of pay.
D	Confusion set	Treats Java as pass-by-reference for primitives.

2. Answer A Concept 1 • transfer

A method `void birthday(Person p) { p.setAge(p.getAge() + 1); }` is called with a Person object whose age is 30. After the call, what does the caller's `person.getAge()` return?

Target word: effect of mutating an object through a copied reference **Decides:** The copied reference points to the same object, and `setAge` changes that object.

A	Correct	The reference is copied, but both copies point to one object; the mutator changes it.
B	Wrong stem number	The increment from the stem, not the new age.
C	Inverted operation	Applies the primitive rule; the object itself was changed, not the reference reassigned.
D	Adjacent category	Java copies the reference value, not the object.

3. Answer D Concept 2

An inventory app splits 17 pallets evenly across 4 trucks with `int perTruck = 17 / 4;` ~~State rules let each truck carry up to 20 pallets.~~ What value is stored in `perTruck`?

Target word: result of int division **Decides:** Both operands are int, so the fraction is truncated.

A	Camouflage	The truck limit does not decide the arithmetic.
B	Adjacent category	Floating-point result; an int cannot hold a fraction.
C	Confusion set	Rounds up; Java integer division truncates toward zero.
D	Correct	$17 / 4 = 4.25$, truncated to 4 in integer division.

4. Answer C Concept 2 • transfer

Given `int total = 9` and `int count = 2`, which expression makes double `avg` equal 4.5?

Target word: where the cast must go **Decides:** An operand must be double before the division happens.

A	Confusion set	Rounds an already truncated 4; returns 4.
B	Inverted operation	Divides first (4) and then casts, giving 4.0.
C	Correct	<code>total</code> becomes 9.0 first, so $9.0 / 2 = 4.5$.
D	Dropped element	$9 / 2$ is already 4 before the multiply; result 4.0.

5. Answer C Concept 3

A login service ~~written by an outside contractor~~ checks if (inputRole == storedRole). Both variables hold the text ADMIN, but inputRole was built at runtime from keyboard input. ~~The service handles about 3,000 logins a day.~~ Why does the check return false?

Target word: why two equal-looking Strings compare false **Decides:** == compares references; the runtime String is a different object.

A	Camouflage	Login volume has nothing to do with the comparison.
B	Absolute language	Two references to the same interned literal are == equal.
C	Correct	Two different String objects with the same text are not == equal.
D	Adjacent category	Both hold ADMIN in the same case; case is not the issue.

6. Answer C Concept 3 · transfer

Which expression correctly tests whether two int primitives a and b hold the same value?

Target word: correct comparison for primitives **Decides:** Primitives are compared by value with ==.

A	Confusion set	Assignment, not comparison.
B	Inverted operation	Applies the object rule; an int has no methods, so this does not compile.
C	Correct	For primitives, == compares the stored values.
D	Adjacent category	Returns 1 only when a is greater than b; 0 means equal.

7. Answer B Concept 4

A billing class already has double total(double price, int qty). ~~Working against a Friday release,~~ a developer adds int total(double price, int qty) to the same class. What happens?

Target word: whether the new method is a legal overload **Decides:** The parameter lists are identical; only the return type differs.

A	Adjacent category	Overriding happens between a parent class and a subclass, not inside one class.
B	Correct	Overloads must differ in number, type, or order of parameters.
C	Absolute language	Same-name methods are legal when their parameter lists differ.
D	Confusion set	Java does not choose overloads by return type.

8. Answer A Concept 5

A Ticket class declares private static int issued = 0; and its constructor runs issued++. A theater box office ~~that has been open since 1998~~ creates three Ticket objects ~~for a Sunday matinee~~. What is the value of issued afterward?

Target word: value of a shared static counter **Decides:** One static copy is incremented by each of three constructors.

A	Correct	All objects share the single static field; it is incremented three times.
B	Confusion set	Treats the static field as an instance field.
C	Camouflage	The year the theater opened does not decide the counter.
D	Adjacent category	Confuses static with final.

9. Answer A Concept 6

A BankAccount class stores balance as a public double field. The security team, ~~which reviews code on Tuesdays~~, flags that other classes can set balance to a negative number. Which change best applies encapsulation?

Target word: best encapsulation fix **Decides:** Outside code must be blocked from direct writes and forced through validation.

A	Correct	Private hides the field; the methods enforce the rules.
B	Adjacent category	Protected is still visible to the package and subclasses, with no validation.
C	Inverted operation	Final blocks all changes, including valid deposits.
D	Dropped element	Leaves out the enforcement that encapsulation requires.

10. Answer D Concept 7

Class Shape has double area() that returns 0. Class Circle extends Shape and overrides area() to return $3.14 * r * r$. A drawing tool ~~used by 400 architects~~ runs Shape s = new Circle(2); System.out.println(s.area()); What prints?

Target word: which area() runs **Decides:** The runtime object is a Circle, so Circle's override runs.

A	Dropped element	3.14×2 drops the second r.
B	Absolute language	A Shape reference can always hold a subclass object and call area().
C	Confusion set	Uses the declared type Shape to choose the method.
D	Correct	Dynamic dispatch runs Circle.area(): $3.14 \times 2 \times 2 = 12.56$.

11. Answer D Concept 7 · transfer

Shape s = new Circle(2); is followed by s.getRadius(); where getRadius() is defined only in Circle, not in Shape. What happens?

Target word: which methods the declared type allows **Decides:** The compiler checks calls against the declared type Shape.

A	Inverted operation	Applies the runtime-dispatch rule to method visibility, which is checked at compile time.
B	Adjacent category	That exception comes from a bad cast; this never compiles.
C	Confusion set	Java has no default value for a missing method.
D	Correct	The declared type limits which methods may be called; a cast to Circle is needed.

12. Answer B Concept 8

A Product class ~~written for a grocery chain with 60 stores~~ declares exactly one constructor, public Product(String sku, double price). Another class calls new Product(). What happens?

Target word: whether a no-argument constructor exists **Decides:** A constructor is declared, so Java adds no default.

A	Adjacent category	The problem is caught at compile time, not runtime.
B	Correct	Declaring any constructor removes the automatic default constructor.
C	Confusion set	Describes default field values, which would apply only if the call compiled.
D	Absolute language	Only when the class declares no constructors.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Java is pass-by-value	Java copies the argument into the parameter; reassigning a primitive or a reference inside a method never changes the caller's variable, but calling a mutator through a copied reference changes the shared object.	Believing Java passes primitives by reference, or that object changes never reach the caller.	A method changes its parameter and the caller prints the original variable afterward.
2	Integer division and casting	int / int yields an int with the fraction truncated; cast an operand to double before dividing to keep the fraction.	Expecting 7 / 2 to be 3.5, or casting the result after the division is already done.	Two int variables divided, often stored into a double.
3	== versus .equals()	== compares primitive values or object references; .equals() compares object content such as String text.	Using == to compare two Strings built at different times.	Two Strings that look the same but one came from input or concatenation.
4	Overloading rules	Overloaded methods share a name but must differ in parameter list; return type alone does not distinguish them.	Thinking a different return type creates a valid overload.	Two methods with the same name and the same parameter types.
5	Static versus instance members	A static field has one copy shared by every object of the class; an instance field has one copy per object.	Assuming each new object gets its own copy of a static counter.	The keyword static on a field changed in a constructor.
6	Encapsulation with access modifiers	Make fields private and expose controlled public methods that validate changes.	Thinking protected or final gives the same protection as private plus validation.	Other classes can set a field to an invalid value.
7	Polymorphism and dynamic dispatch	The runtime object type decides which overridden method runs; the declared reference type decides which methods can be called at all.	Running the parent's version because the variable is declared as the parent type.	Parent p = new Child(); followed by a method call.
8	Constructors and the default constructor	Java supplies a no-argument default constructor only when a class declares no constructors at all.	Believing a no-argument constructor always exists.	A class with one parameterized constructor is created with new ClassName().

Course scope (public description): Java Fundamentals introduces you to object-oriented programming in the Java language. You will create and call methods, design Java classes, and other object-oriented principles and constructs to develop software that meets business requirements. This course requires foundational knowledge of programming including variables, type, program flow and debugging.